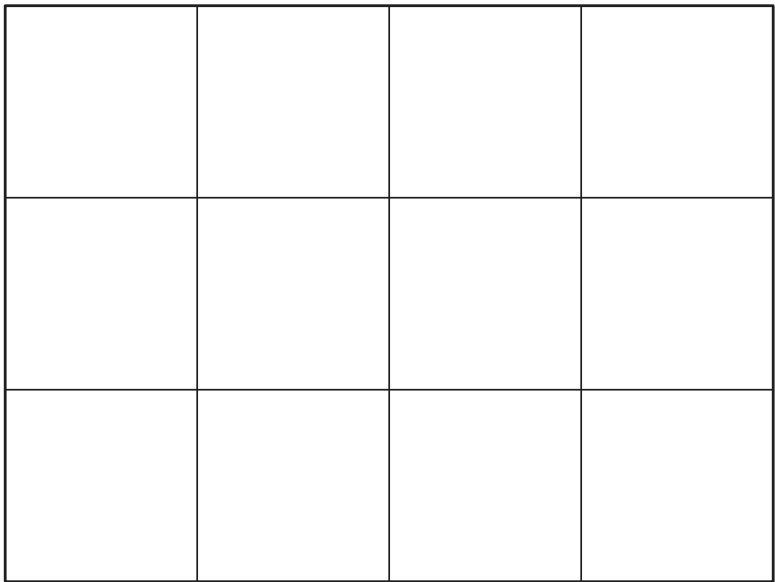
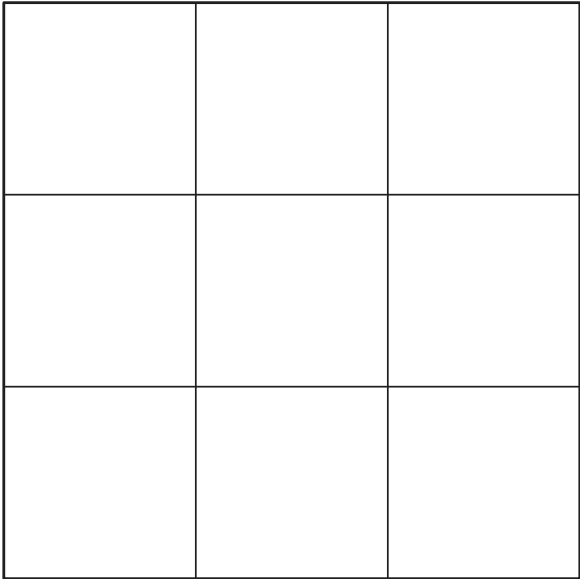
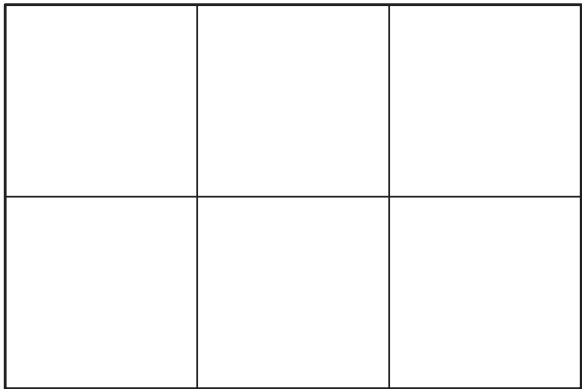
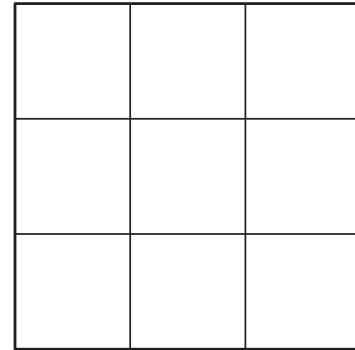
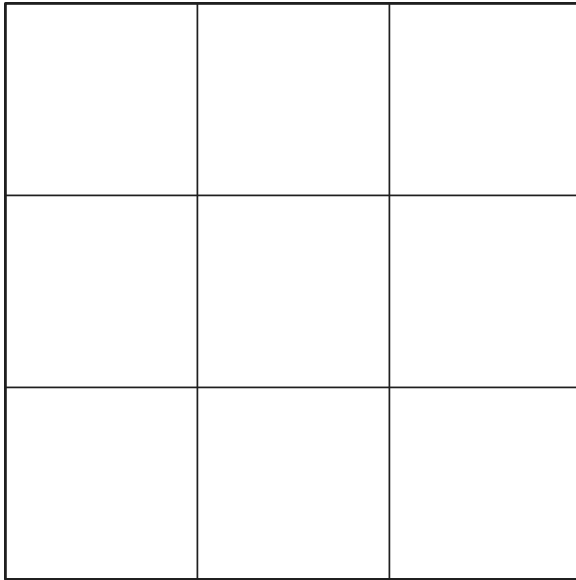


Problem 1: A domino covers two squares. Cover the 2-by-3 board with dominoes: every square covered, no overlaps, nothing sticking out. Then try to cover the 3-by-3 board and the 3-by-4 board. For each board, write in the table how many squares it has, whether you covered it (yes or no), and how many dominoes you used or how many squares you could not cover. Then, without building, predict yes or no for a 3-by-5 board, a 4-by-4 board and a 5-by-5 board, and write your predictions in the table. Check one prediction on the grid paper.

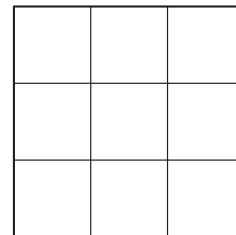
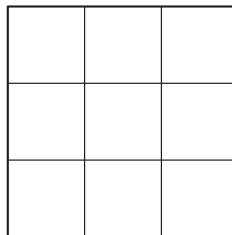
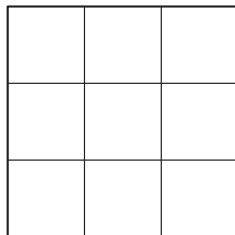
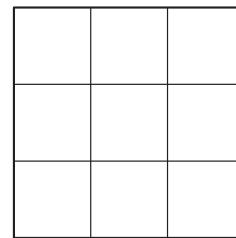
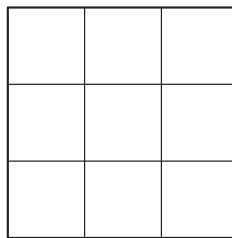
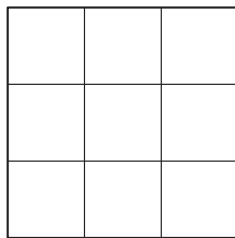


	squares	covered? yes / no	dominoes used, or squares left
2-by-3			
3-by-3			
3-by-4			
3-by-5 (prediction)			
4-by-4 (prediction)			
5-by-5 (prediction)			

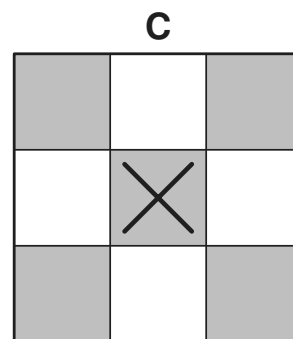
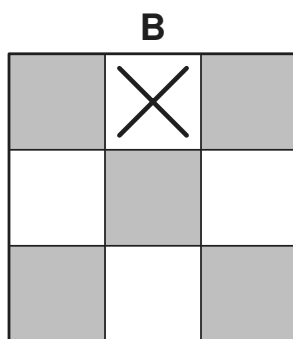
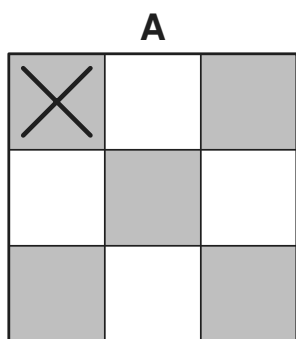
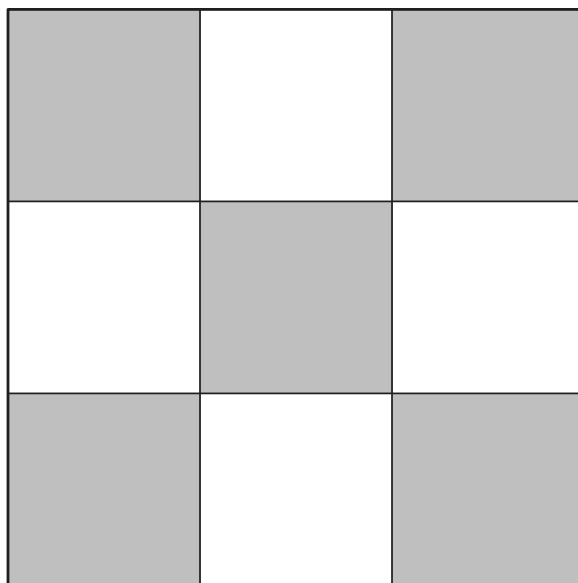
Problem 2: Put the X square on one square of the 3-by-3 board. That square is crossed out and may not be covered. Try to cover the other eight squares with four dominoes. In the result grid, draw a tick in the square where the X was if you covered the board, and a cross if you could not. Do this for all nine squares. Draw each different cover you found on a small board. Look at the pattern of ticks and crosses.



result grid

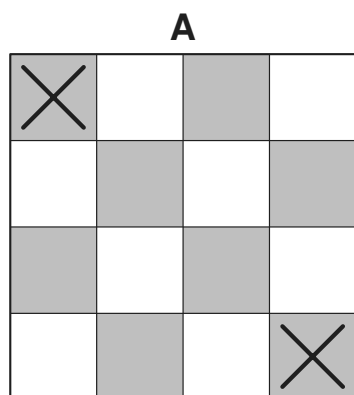
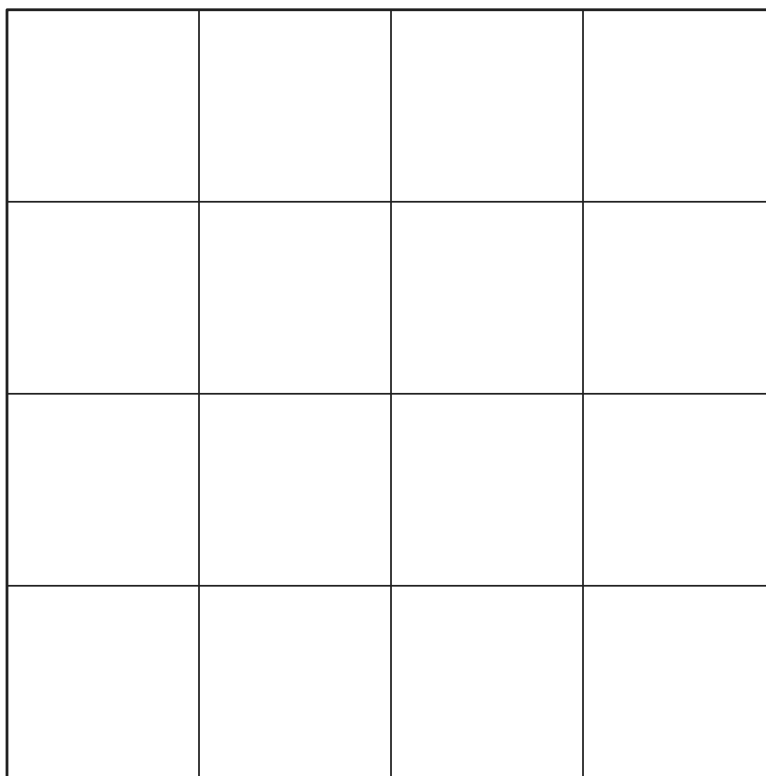


Problem 3: This 3-by-3 board is shaded like a checkerboard. Put one domino anywhere on it and look at the colours of the two squares under it. Move it to five other places and look each time. Now count the dark squares and the light squares that are not crossed out on boards A, B and C, and write the numbers in the table. In the last column, write yes or no: was that board covered in Problem 2?



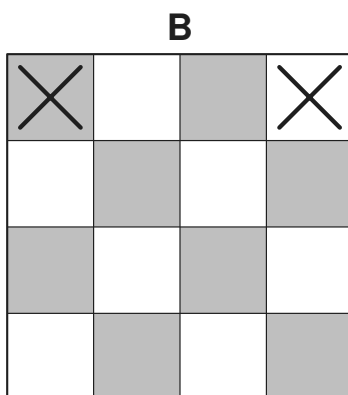
	dark	light	covered in Problem 2?
A			
B			
C			

Problem 4: Put two X squares on the 4-by-4 board on the corners shown in picture A. Cover the other 14 squares with seven dominoes, or decide that it cannot be done. Draw your cover on picture A, or cross picture A out. Do the same for pictures B and C. Under each picture, write how many dark squares and how many light squares are not crossed out.



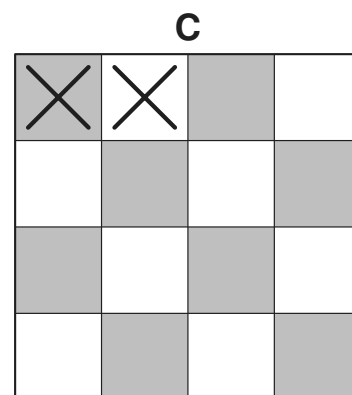
dark _____

light _____



dark _____

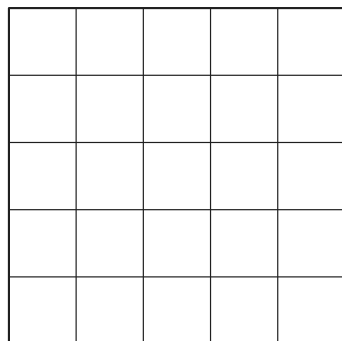
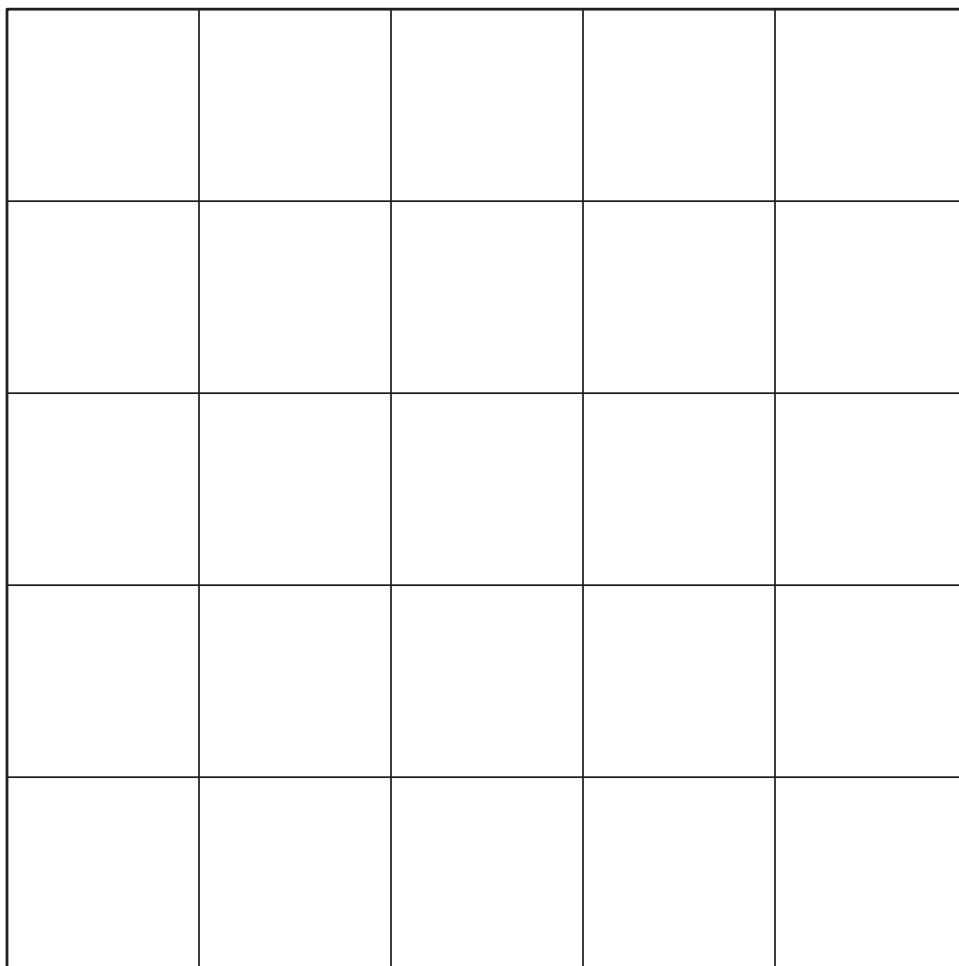
light _____



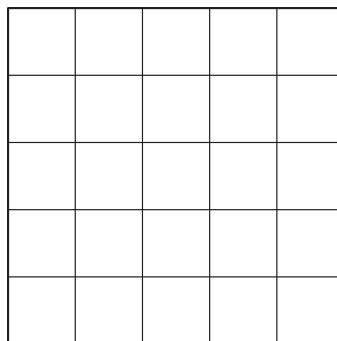
dark _____

light _____

Problem 5: Before you place anything: on the prediction grid, mark every square that you think can be crossed out so that 12 dominoes cover the rest of the 5-by-5 board. Then test at least four of your marked squares and two unmarked squares: put the X square there and try to cover the rest. Draw a tick or a cross in the result grid each time. Compare the two grids.

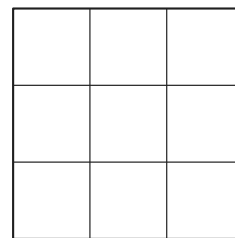
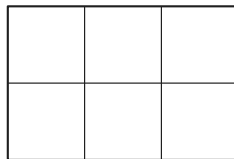
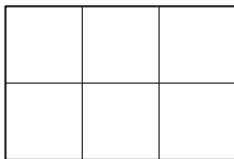
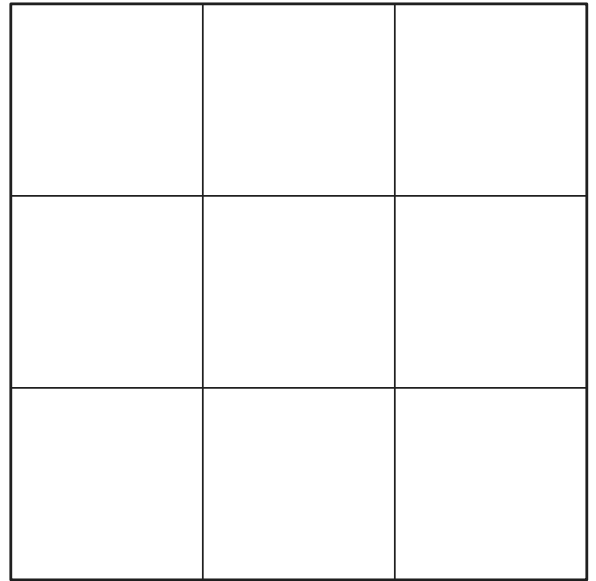
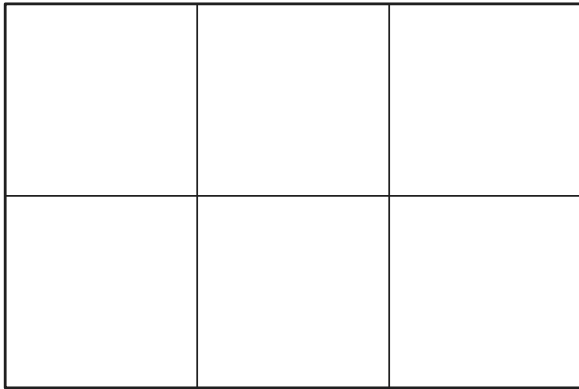


prediction grid

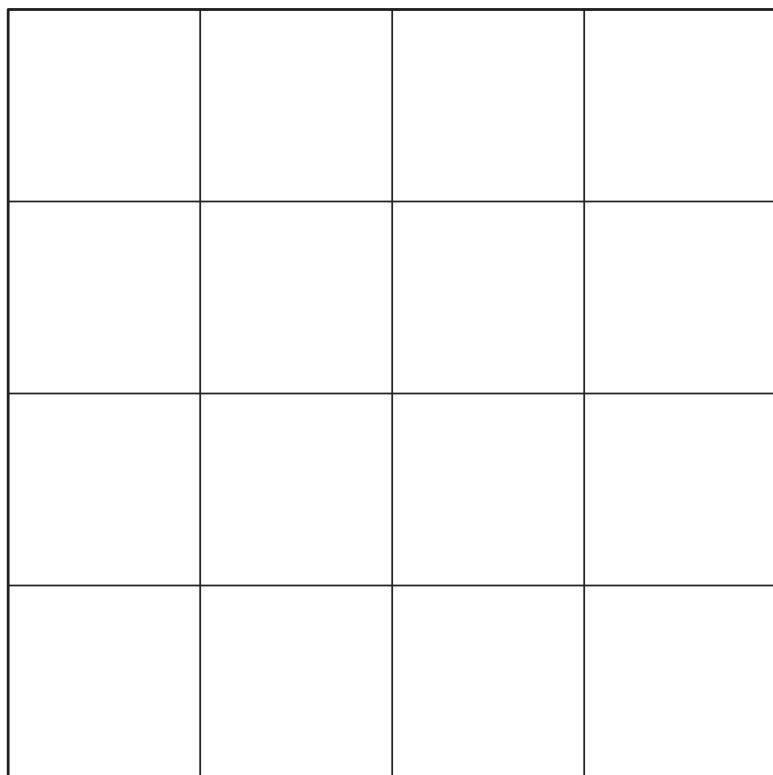
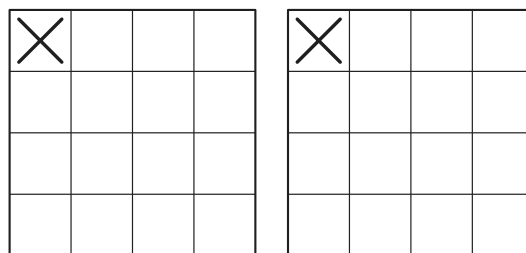
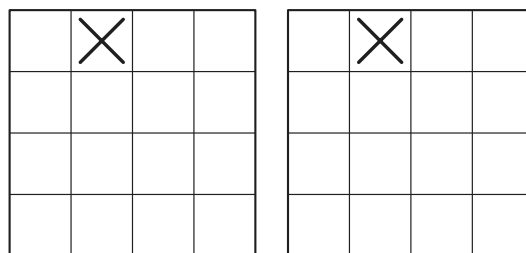
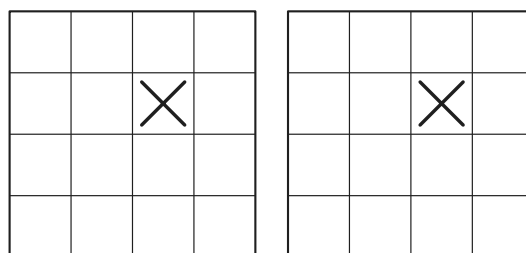


result grid

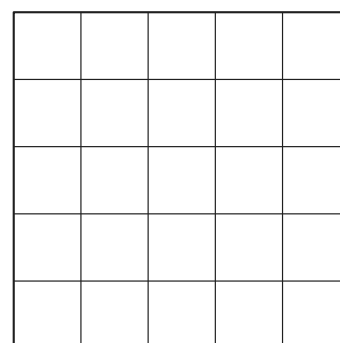
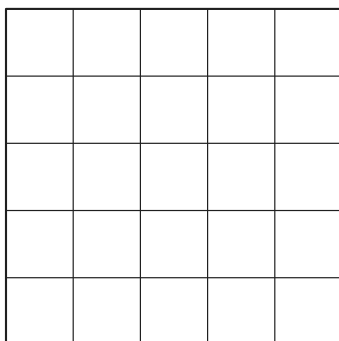
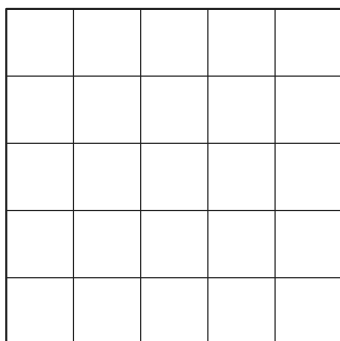
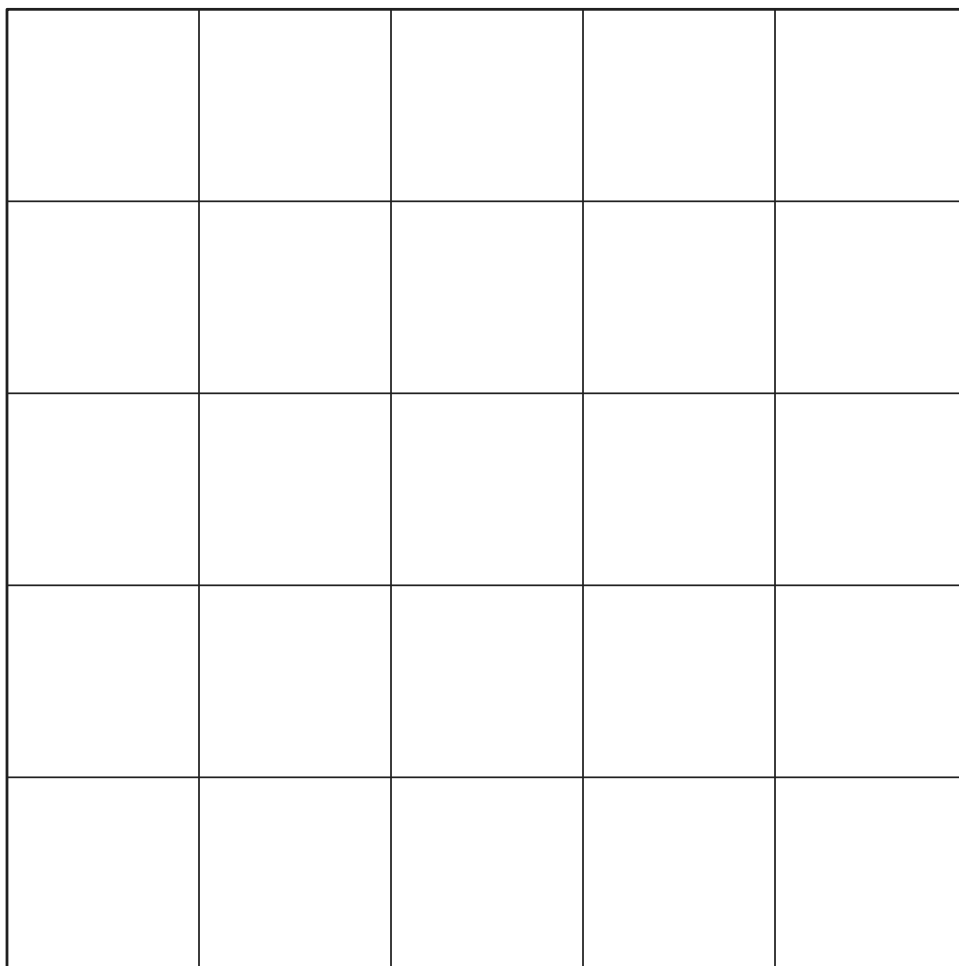
Problem 6: An L-piece covers three squares in an L shape. Cover the 2-by-3 board with two L-pieces and draw your cover on the first small board. Do not turn the board. Find a different cover and draw it on the second small board. Now try to cover the 3-by-3 board with three L-pieces. If you can, draw it on the third small board. If you cannot, cross out the third small board and tell the adult why.



Problem 7: Put the X square on the corner of the 4-by-4 board shown in picture A. Cover the other 15 squares with five L-pieces and draw your cover on picture A. Then look for a second cover of the same board that looks different, without turning the board. Draw it on the second picture A, or cross that picture out if you cannot find one. Do the same for the edge square in picture B and the inside square in picture C.

**A****B****C**

Problem 8: Put the X square on the centre square of the 5-by-5 board and cover the other 24 squares with eight L-pieces. Draw the cover on the first small board. Now put the X square on a corner and try again. Draw the cover on the second small board. Then try the middle square of the top edge, and then the square next to the corner along the top edge. Each time, draw a tick or a cross in the result grid in the square where the X was. If there is time, try more squares.



result grid